MILITARY SPECIFICATION SHEET

# PRINTED CIRCUIT CARDS, M1 AND M1A1 TANK DRIVER'S INSTRUMENT PANEL, DIP-A1

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the Driver's Instrument Panel, DIP-Al, described herein shall consist of this document and the latest issue in effect of specification MIL-P-62341.

## 3. REQUIREMENTS

3.4 Design and construction. The printed circuit card (PCC) DIP-A1 provides switching circuits for monitoring and maintenance functions.

3.4.1 Configuration.

3.4.1.1 Class 1. Printed circuit card DIP-A1 shall be fabricated and asembled in accordance with Drawing 12282607.

3.4.1.2 Class 2. Printed circuit card DIP-A1 shall be fabricated and assembled in accordance with Drawing 12337028.

3.4.2 Weight. The weight of the PCC shall not exceed 1.0 pound (lb) (see 4.8.3).

3.4.4 Nuclear hardening. Not applicable.


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3.5.3 Diode I/V characteristics.

3.5.3.1 Forward voltage drop. The forward voltage drop of the diodes shall not exceed 0.9 V dc at 0.5 A when measured individually as specified in table I or II (see 4,8.4.4.1).

3.5.3.2 Reverse leakage current. The leakage current in the reverse direction shall not exceed 50 microamperes when measured as specified in table I or 11 (see 4.8.4.4.2).

3.5.3.3 Class 1.

TABLE I. Forward and reverse voltage.

| Diode (CR) | Forward voltage | | Reverse voltage | | Remarks |
|---|---|---|---|---|---|
| | Positive | Return | Positive | Return | |
| 16 | P1-2 | P1-12 | P1-12 | P1-2 | |
| 27 | P1-4 | P1-12 | P1-12 | P1-4 | |
| 25 | P1-6 | P1-12 | P1-12 | P1-6 | |
| 13 | P1-9 | P1-12 | P1-12 | P1-9 | |
| 24 | P1-11 | P1-12 | P1-12 | P1-11 | |
| 11 | P1-13 | P1-12 | P1-12 | P1-13 | |
| 23 | P1-18 | P1-12 | P1-12 | P1-18 | |
| 9 | P1-30 | P1-12 | P1-12 | P1-30 | |
| 19 | P1-33 | P1-26 | P1-26 | P1-33 | |
| 10 | P1-19 | P1-26 | P2-26 | P1-19 | |
| 12 | P1-15 | P1-26 | | | Energize K5 (P1-15+/P1-10-) |
| 15 | P1-15 | P1-26 | | | Energize K6 (P1-15+/P1-8-) |
| 18 | P1-15 | P1-26 | | | Energize K7 (P1-15+/P1-1-) |
| 14 | P1-15 | P1-26 | | | Energize K8 (P1-15+/P1-5-) |
| 17 | P1-15 | P1-26 | | | Energize K9 (P1-15+/P1-3-) |
| 30 | P1-31 | P1-26 | | | Energize K14 (P1-31+/P1-17-) |



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3.5.3.4 Class 2.

TABLE II. Forward and reverse voltage.

| Diode (CR) | Forward voltage | | Reverse voltage | | Remarks |
|---|---|---|---|---|---|
| | Positive | Return | Positive | Return | |
| 16 | P1-57 | P1-36 | P1-36 | P1-57 | |
| 27 | P1-59 | P1-36 | P1-36 | P1-59 | |
| 25 | P1-41 | P1-36 | P1-36 | P1-41 | |
| 13 | P1-18 | P1-36 | P1-36 | P1-18 | |
| 24 | P1-37 | P1-36 | P1-36 | P1-37 | |
| 11 | P1-35 | P1-36 | P1-36 | P1-35 | |
| 23 | P1-30 | P1-36 | P1-36 | P1-30 | |
| 9 | P1-45 | P1-36 | P1-36 | P1-45 | |
| 19 | P1-44 | P1-24 | P1-24 | P1-44 | |
| 10 | P1-29 | P1-24 | P1-24 | P1-29 | |
| 12 | P1-14 | P1-24 | | | Energize K5 (P1-14+/P1-38-) |
| 15 | P1-14 | P1-24 | | | Energize K6 (P1-14+/P1-39-) |
| 18 | P1-14 | P1-24 | | | Energize K7 (P1-14+/P1-56-) |
| 14 | P1-14 | P1-24 | | | Energize K8 (P1-14+/P1-60-) |
| 17 | P1-14 | P1-24 | | | Energize K9 (P1-14+/P1-58-) |
| 30 | P1-46 | P1-24 | | | Energize K14 (P1-46+/P1-16-) |
| 31 | P1-46 | P1-24 | | | Energize K13 (P1-2+/P1-23-) |
| 32 | P1-14 | P1-58 | | | |
| 35 | P1-27 | P1-36 | P1-36 | P1-27 | |
| 36 | P1-33 | P1-32 | P1-32 | P1-12 | |
| 38 | P1-14 | P1-24 | | | Energize K10 (P1-14+/P1-25-) |
| 39 | P1-32 | P1-36 | P1-36 | P1-32 | |
| 40 | P1-10 | P1-42,22 | P1-42,22 | P1-10 | |

3.5.4 Blocking diodes. Not applicable.

3.5.5 Resistors. Not applicable.

3.5.6 Relay operation. Relays shall be energized by applying 24 ± 6 V dc to the terminals defined in table III or IV (see 4.8.4.7).



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3.5.6.1 Class 1.

TABLE III. Relay operation.

| Relay | Input Terminals Contact | | Continuity |
|---|---|---|---|
| | Positive | Return | |
| K1 | P1-34 | P1-20 | P1-31 to P1-15 |
| K2 | P1-29 | P1-20 | P1-7 to P1-30 |
| K3 | P1-35 | P1-20 | P1-7 to P1-30 |
| K4 and K2 | P1-14 and 29 | P1-20 | P1-15 to P1-19 |
| K4 and K3 | P1-14 and 35 | P1-20 | P1-15 to P1-19 |
| K5 | P1-15 | P1-10 | P1-7 to P1-13 |
| K6 | P1-15 | P1-8 | P1-7 to P1-9 |
| K7 | P1-15 | P1-1 | P1-7 to P1-2 |
| K8 | P1-15 | P1-5 | P1-7 to P1-6 |
| K9 | P1-15 | P1-3 | P1-7 to P1-4 |
| K12 | P1-21 | P1-20 | P1-22 to P1-23 |

3.5.6.2 Class 2.

TABLE IV. Relay operation.

| Relay | Input Terminals Contact | | Resistance | | Measure (ohms) |
|---|---|---|---|---|---|
| | Positve | Return | From | To | |
| K1 | P1-47 | P1-28 | P1-46 | P1-14 | 1.0 |
| K2 | P1-43 | P1-28 | P1-40 | P1-45 | 1.0 |
| K3 | P1-21 | P1-28 | P1-40 | P1-45 | 1.0 |
| K4 and K2 | P1-34 and 43 | P1-28 | P1-14 | P1-45 | 1.0 |
| K4 and K3 | P1-34 and 21 | P1-28 | P1-14 | P1-29 | 1.0 |
| K5 | P1-14 | P1-38 | P1-40 | P1-35 | 1.0 |
| K6 | P1-14 | P1-39 | P1-40 | P1-18 | 1.0 |
| K7 | P1-14 | P1-56 | P1-40 | P1-57 | 1.0 |
| K8 | P1-14 | P1-60 | P1-40 | P1-41 | 1.0 |
| K9 | P1-14 | P1-58 | P1-40 | P1-59 | 1.0 |
| K10 | P1-14 | P1-8 | P1-40 | P1-27 | 1.0 |
| K12 | P1-14 | P1-28 | P1-26 | P1-25 | 1.0 |
| K15 | P1-14 | P1-6 | P1-10 | P1-42 | 1.0 |

3.5.7 Transient voltage characteristics.

3.5.7.1 Class 1.

3.5.7.1.1 Internal. Unless otherwise specified, upon removal of 24 ± 6 V dc from relay mils K1 through K14 as specified in table V, the relay mil dropout transient voltage shall be within the range of ± 30 V dc (see 4.8.4.8.1.1).



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3.5.7.1.2 External. With 24 ± 6 V dc applied across three relay mils (conforming to MIL-R-5757/23) connected in parallel to pins P1-33 (positive) and P1-20 (return), the dropout transient voltage shall be within the range of ± 30 V dc upon removal of 24 + 6 V dc from the relay mils (see 4.8.4.8.1.2).

TABLE V. Relay transient voltage suppression

| Circuit | Voltage | | Remove Power From | Remarks |
|---|---|---|---|---|
| | Positive | Return | | |
| K1 | P1-34 | P1-20 | P1-34 | CK's CR-1/VR-1 |
| K2 | P1-29 | P1-20 | P1-29 | CK's CR-2/VR-2 |
| K3 | P1-35 | P1-20 | P1-35 | CK's CR-4/VR-3 |
| K4 | P1-14 | P1-20 | P1-14 | CK's CR-5/VR-4 |
| K5 | P1-15 | P1-10 | P1-10 | CK's CR-6/VR-5 |
| *K6 | P1-15 | P1-8 | P1-8 | (CK's CR-7) |
| K7 | P1-15 | P1-1 | P1-1 | CK's CR-8/VR-7 |
| K8 | P1-15 | P1-5 | P1-5 | CK's CR-26/VR-10 |
| *K9 | P1-15 | P1-3 | P1-3 | (CK's CR-28) |
| K10,K11,K13 | P1-32 | P1-27 | P1-32 | CK's CR-21/VR-9 |
| K12 | P1-21 | P1-20 | P1-21 | CK's CR-3 |
| K14 | P1-31 | P1-20 | P1-31 | CK's CR-29/VR-12 |

*Drop out transient voltage shall be within the range of +30 to -0.9 V dc.

3.5.7.2 Class 2.

3.5.7.2.1 Internal. Unless otherwise specified, upon removal of 24 ± 6 V dc from relay mils K1 through K14 as specified in table VI, the relay mil dropout transient voltage shall be within the range of ± 30 V dc (see 4.8.4.8.2.1).

3.5.7.2.2 External. With 24 ± 6 V dc applied across three relay mils (conforming to MIL-R-5757/23) connected in parallel to pins P1-44 (positive) and P1-28 (return), the dropout transient voltage shall be within the range of ± 30 V dc upon removal of 24 ± 6 V dc from the relay mils (see 4.8.4.8.2.2).

3.5.8 Time delay and latching circuit. Not applicable.



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TABLE VI. Relay transient voltage suppression.

| Circuit | Voltage | | Remove Power From | Remarks |
|---|---|---|---|---|
| | Positive | Return | | |
| K1 | P1-47 | P1-28 | P1-47 | CK's CR-1/VR-1 |
| K2 | P1-43 | P1-28 | P1-43 | CK's CR-2/VR-2 |
| K3 | P1-21 | P1-28 | P1-21 | CK's CR-4/VR-3 |
| K4 | P1-34 | P1-28 | P1-34 | CK's CR-5/VR-4 |
| K5 | P1-14 | P1-38 | P1-38 | CK's CR-6/VR-5 |
| *K6 | P1-14 | P1-39 | P1-39 | (CK's CR-7) |
| K7 | P1-14 | P1-56 | P1-56 | CK's CR-8/VR-7 |
| K8 | P1-14 | P1-60 | P1-60 | CK's CR-26/VR-10 |
| *K9 | P1-14 | P1-58 | P1-58 | (CK's CR-28) |
| K11,K13 | P1-2 | P1-23 | P1-2 | CK's CR-21/VR-9 |
| K12 | P1-27 | P1-28 | P1-27 | CK's CR-3 |
| K14 | P1-46 | P1-28 | P1-46 | CK's CR-29/VR-12 |

*Drop out transient voltage shall be within the range of +30 to -0.9 V dc.

3.5.9 Fire warning circuit.

3.5.9.1 Class 1. With 24 ±6 V dc connected to pins P1-31 (positive) and P1-27 (return), connect a test lamp (conforming to MS25237-387) to pins P1-16 (positive) and P1-18 (return). Connect a second test lamp to pins P1-26 (positive) and P1-25 (return), with a 1.0 pulse per second (pps) repetitive positive signal source having an amplitude of 24 ± 6 V dc connected to pins P1-32 (positive) and P1-27 (return), relays K10, K11 and K13 shall latch in and cause the test lamps to flash at the signal source rate of 1.0 pps. Relays K10, K11 and K13 shall not de-energize when the signal source rate of 1.0 pps. Relays K10, K11 and K13 shall not de-energize when the signal source is removed from pin P1-32. Relays K10, K11 and K13 shall deenergize with a momentary interruption of power at pin P1-31 (see 4.8.4.10.1).

3.5.9.2 Class 2. With 24 ± 6 V dc connected to pins P1-46 (positive) and P1-23 and P1-26 (return), connect a test lamp (conforming to MS25237-387) to pins P1-31 (positive) and P1-30 (return). Connect a second test lamp to pins P1-24 (positive) and P1-8 (return). With a 1.0 pulse per second (pps) repetitive positive signal source having an amplitude of 24 ± 6 V dc connected to pins P1-2 (positive) and P1-23 (return), the test lamp between P1-24 and P1-8 shall illuminate steady and the test lamp between P1-31 and P1-30 shall flash at the signal source rate of 1.0 pps. When the signal source is removed from P1-2, the lamp between P1-24 and P1-8 shall remain illuminated and the flashing lamp between P1-31 and P1-30 shall extinguish. The lamp between P1-24 and P1-8 shall extinguish with a momentary interruption of power at P1-46 (see 4.8.4.10.2).



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4.8 Methods of inspection.

4.8.3 Weight. Conformance to 3.4.2 shall be determined by weighing the printed circuit card on a scale.

4.8.4.4 Diode I/V circuit characteristics.

4.8.4.4.1 Forward voltage drop. To determine conformance to 3.5.3.1, the forward voltage drop of the diodes as specified in table I or II shall be tested individually in accordance with MIL-STD-750, method 4011.4, at terminal parts listed in table I or II.

4.8.4.4.2 Reverse leakage current. To determine conformance to 3.5.3.2, the reverse leakage current of the diodes as specified in table I or II shall be tested individually in accordance with MIL-STD-750, method 416.3, at terminal points listed in table I or II.

4.8.4.5 Blocking diodes. Not applicable.

4.8.4.6 Resistors. Not applicable.

4.8.4.7 Relay operation. To determine conformance to 3.5.6, using the power supply, apply 24 ± 6 V dc individually to each relay input set of terminals as specified in table III or IV. Using digital multimeter, verify that the energized relay contact resistance, measured at the corresponding continuity terminals specified in table 111 or IV meet the requirements of 3.5.6.

4.8.4.8 Transient Voltage characteristics.

4.8.4.8.1 Class 1.

4.8.4.8.1.1 Internal. To determine conformance to 3.5.7.1.1, utilizing a power supply and an oscilloscope, apply and then remove 24 ± 6 V dc at individual relay circuit terminals as shown in table V. Verify that each circuit voltage transient does not exceed the requirements of 3.5.7.1.1 as indicated on the oscilloscope.

4.8.4.8.1.2 External. To determine conformance to 3.5.7.1.2, assemble three (3) relay coils (conforming to MIL-R-5757/23) in parallel and connect them to the assembly pins P1-33 (positive) and P1-20 (return). Using a power supply and oscilloscope of 4.1.4.2, apply 24 ± 6 V dc through the three (3) coils to pins P1-33 (positive) and P1-20 (return). Remove the voltage from the relay coils and verify that the dropout transient voltage as indicated on the oscilloscope does not exceed the requirements of 3.5.7.1.2.



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4.8.4.8.2 Class 2.

4.8.4.8.2.1 Internal. To determine conformance to 3.5.7.2.1, utilizing a power supply and an oscilloscope, apply and then remove 24 + 6 V dc at individual relay circuit terminals as shown in table VI. Verify that each circuit voltage transient, as indicated on the oscilloscope, does not exceed the requirements of 3.5.7.2.1.

4.8,4.8.2.2 External. To determine conformance to 3.5.7.2.2, assemble three (3) relay mils (conforming to MIL-R-5757/23) in parallel and connect them to the assembly pins P1-44 (positive) and P1-28 (return). Using a power supply and oscilloscope, apply 24 + 6 V dc through the three (3) mils to pins P1-44 (positive) and P1-28 (return). Remove the voltage from the relay coils and verify that the dropout voltage as indicated on the oscilloscope does not exceed the requirements of 3.5.7.2.2.

4.8.4.9 Time delay and latching circuit. Not applicable.

4.8.4.10 Fire warning circuit.

4.8.4.10.1 Class 1. To determine conformance to 3.5.9.1, interconnect two test lamps (conforming to MS25237-387) to the printed circuit card. Connect one lamp to P1-16 (positive) and P1-18 (return) and connect the second lamp to P1-26 (positive) and P1-25 (return). Apply 24 ± 6 V dc between P1-31 (positive) and P1-27 (return) and also supply a 1.0 pulse per second, repetitive positive signal of 24 ± 6 V dc amplitude between P1-32 (positive) and P1-27 (return). Use an electric timer and multimeter to verify that both test lamps flash at the 1.0 pps rate. Interruption of the 1.0 pps signal applied to P1-32 will not deenergize the relays. Verify that relays K10 and K11 are latched by a resistance measurement of 1.0 ohm maximum through closed relay contacts interconnected between P1-25 and P1-27 and between P1-27 and P1-28. The momentarily interruption of 24 ± 6 V dc applied to P1-31 will deenergize relay K13 and simulataneously deenergize relay K10 and K11 resulting a resistance measurement in excess of 100 000 ohms between P1-25 and P1-27 and between P1-27 and P1-18.

4.8.4.10.2 Class 2. To determine conformance to 3.5.9.2, interconnect two test lamps (conforming to MS25237-387) to the assembly. Connect one lamp to P1-31 (positive) and P1-30 (return) and connect the second lamp to P1-24 (positive) and P1-8 (return). Apply 24 ± 6 V dc between P1-46 (positive) and P1-23 (return) and P1-6 (return), and also apply a 1.0 pulse per second, repetitive positive signal of 24 ± 6 V dc amplitude between P1-2 (positive) and P1-23 (return). Use an electric timer and multimeter to verify that the test lamp between P1-31 and P1-30 flashes at the 1.0 pps rate and that the test lamp between P1-24 and P1-8 illuminates steady. Interruption of the 1.0 pps signal applied ot P1-2 shall cause the flashing lamp between P1-31 and P1-30 to extinguish while the lamp between P1-24 and P1-25 remains illuminated. The momentary interruption of 24 ±6 V dc applied to P1-46 shall cause the lamp between P1-24 and P1-8 to extinguish.

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